THE GHOST LEMMA: A COMPOSITION OF MAPS THAT ARE ZERO ON COHOMOLOGY BETWEEN STAGES OF THE GENERATED SUBCATEGORY VANISHES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 20663-20677 (canonical tag 0FXH) and the counted proof is the proof environment at source lines 20679-20708, which immediately follows it and is its own proof.

Editorial note. The counted statement contains the author's reference environment recording the origin of the result; the retained preamble defines that environment as comment-type, so its body and the citation inside it are discarded when the document is typeset, which is why the card needs no bibliography.

Editorial note. The two references are to the section introducing the notation of the stages and to the representability statement used in the induction; the kinds recorded are the type words appearing in the labels themselves. Both are navigation stubs with their roles recorded in the reasons.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which is exactly what makes the reference environment a comment; the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- An object of the first stage of the subcategory generated by an object is a direct summand of a finite direct sum of shifts of that object, and maps out of a shift of the ring into a complex are computed by the cohomology of the complex. used at proof lines 20679-20686. Reduces the case of one stage to the vanishing of a map whose induced map on cohomology is zero.
- An object of the n-th stage of the generated subcategory sits in a distinguished triangle whose first term is of the first stage and whose third term is of the previous stage. used at proof lines 20689-20694. Provides the induction step, after replacing the object by the object of the triangle.
- A morphism whose composite with the map to the third term of a distinguished triangle vanishes factors through that third term, by the representability statement for the homological functor. invoked at proof lines 20701-20703. Factors the first map of the chain and reduces the claim to the induction hypothesis applied to the remaining maps.

Source reference. Original section 09SI: sources/stacks-chow/tags.txt, source lines 12356–12356. The reference points to the section where the notation for the stages

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of the generated subcategory is introduced, cited at proof line 20669; it fixes notation and the induction over stages is inside the counted span.

Source reference. Original lemma 0149: sources/stacks-chow/tags.txt, source lines 1464–1464. The reference is applied at proof lines 20701-20703 to factor the first map of the chain through the third term of the distinguished triangle; the induction that consumes the factorization is inside the counted span.

Lemma 0.2. *Let R be a ring. Let $n \geq 1$. Let $K \in \langle R \rangle_n$ with notation as in Derived Categories, Section 09SI. Consider maps*

$$K \xrightarrow{f_1} K_1 \xrightarrow{f_2} K_2 \xrightarrow{f_3} \ldots \xrightarrow{f_n} K_n$$

in $D(R)$. If $H^i(f_j) = 0$ for all i, j, then $f_n \circ \ldots \circ f_1 = 0$.

Proof. If $n = 1$, then K is a direct summand in $D(R)$ of a bounded complex $P^\bullet$ whose terms are finite free R-modules and whose differentials are zero. Thus it suffices to show any morphism $f : P^\bullet \to K_1$ in $D(R)$ with $H^i(f) = 0$ for all i is zero. Since $P^\bullet$ is a finite direct sum $P^\bullet = \bigoplus R[m_j]$ it suffices to show any morphism $g : R[m] \to K_1$ with $H^{-m}(g) = 0$ in $D(R)$ is zero. This follows from the fact that $\mathrm{Hom}_{D(R)}(R[-m], K) = H^m(K)$.

For $n > 1$ we proceed by induction on n. Namely, we know that K is a summand in $D(R)$ of an object P which sits in a distinguished triangle

$$P' \xrightarrow{i} P \xrightarrow{p} P'' \to P'[1]$$

with $P' \in \langle R \rangle_1$ and $P'' \in \langle R \rangle_{n-1}$. As above we may replace K by P and assume that we have

$$P \xrightarrow{f_1} K_1 \xrightarrow{f_2} K_2 \xrightarrow{f_3} \ldots \xrightarrow{f_n} K_n$$

in $D(R)$ with f_j zero on cohomology. By the case $n = 1$ the composition $f_1 \circ i$ is zero. Hence by Derived Categories, Lemma 0149 we can find a morphism $h : P'' \to K_1$ such that $f_1 = h \circ p$. Observe that $f_2 \circ h$ is zero on cohomology. Hence by induction we find that $f_n \circ \ldots \circ f_2 \circ h = 0$ which implies $f_n \circ \ldots \circ f_1 = f_n \circ \ldots \circ f_2 \circ h \circ p = 0$ as desired. $\square$